Problem 10.31

Find an expression for the fraction of a period at which the excess of present values computed at simple discount over compound discount is a maximum.

Hint: See Example 10.13.

Example 10.13

Find an expression of t in terms of δ so that

$$f(t) = (1 + it) - (1 + i)^t$$

is maximum.

Solution to Example 10.13.

Since

$$f'(t) = i - (1 + i)^t \ln(1 + i) = i - \delta(1 + i)^t,$$

we see that $f'(t) = 0$ when

$$(1 + i)^t = \frac{i}{\delta}.$$

Solving for t, we find

$$t = \frac{\ln i - \ln \delta}{\ln(1 + i)} = \frac{\ln i - \ln \delta}{\delta}.$$

Moreover,

$$f''(t) = -\delta^2(1 + i)^t < 0,$$

so the critical point of f is a maximum.

Solution to Problem 10.31.

Let d be the effective discount rate per period. The present value of one unit due at time t is

$$v_{\mathrm{s}}(t) = 1 - dt$$

under simple discount, whereas under compound discount it is

$$v_{\mathrm{c}}(t) = (1 - d)^t.$$

Therefore, the excess of the simple-discount present value over the compound-discount present value is

$$f(t) = (1 - dt) - (1 - d)^t.$$

Recall that the force of interest δ satisfies

$$1 - d = e^{-\delta}, \qquad \ln(1 - d) = -\delta.$$

Differentiating f gives

$$\begin{aligned} f'(t) &= -d - (1-d)^t \ln(1-d) \\ &= -d + \delta(1-d)^t. \end{aligned}$$

At a critical point,

$$\begin{aligned} f'(t) &= 0, \\ \delta(1-d)^t &= d, \\ (1-d)^t &= \frac{d}{\delta}. \end{aligned}$$

Taking logarithms and using $\ln(1-d) = -\delta$, we obtain

$$\begin{aligned} t\ln(1-d) &= \ln\left(\frac{d}{\delta}\right), \\ t &= \frac{\ln(d/\delta)}{-\delta} = \frac{\ln(\delta) - \ln(d)}{\delta}. \end{aligned}$$

Finally,

$$f''(t) = -(1-d)^t\big[\ln(1-d)\big]^2 = -\delta^2(1-d)^t < 0,$$

so this critical point is a maximum. Hence, the required fraction of a period is

$$\boxed{t = \frac{\ln(\delta) - \ln(d)}{\delta}}, \qquad \delta = -\ln(1-d).$$